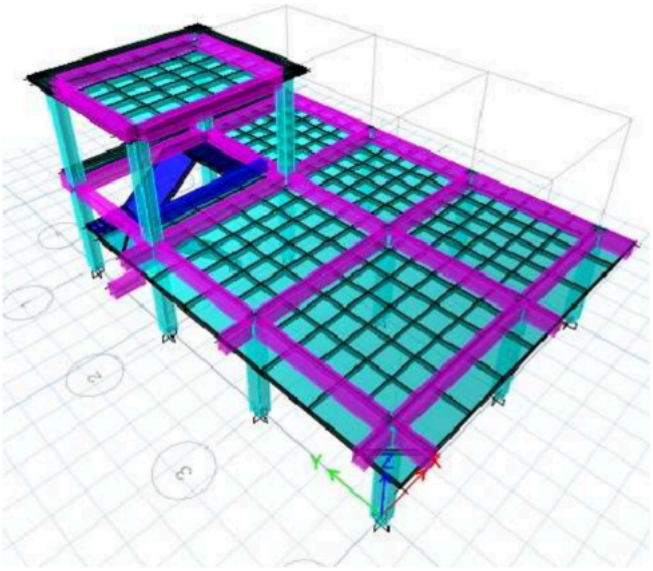
Technical Project Report

DESIGN AND STRUCTURAL ANALYSIS OF A G+1 REINFORCED CONCRETE STRUCTURE

Using AutoCAD 2024 and STAAD.Pro under Seismic Loading



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PREFACE

Structural engineering combines engineering judgement, drafting accuracy and analytical verification to ensure that buildings can safely resist gravity and lateral actions. This technical project focuses on modelling a Ground +1 (G+1) reinforced concrete structure using AutoCAD 2024 and studying the structural response of beams, columns and a cantilever member in
STAAD.Pro under seismic loading.

The work follows a practical workflow: establish a building layout, prepare drawings, idealise the structural frame, assign material and section properties, define load cases, perform analysis and review the resulting member forces and displacements. Particular attention is given to the role of seismic actions, which introduce lateral forces and can govern the response of frames even when gravity loads are modest

The report is organised in a format similar to a conventional internship report, with an abstract, objectives, methodology, software workflow, analysis discussion, learning outcomes, conclusion and references.The diagrams included are explanatory schematics, not project-specific construction drawings or STAAD.Pro

Scope note

The project description supplied did not include the actual floor dimensions, member sizes, concrete/steel grades, seismic zone, soil type, load values or STAAD.Pro output files. Therefore, this report explains the modelling and analysis process without inventing numerical results.

Any final design submission should replace the illustrative diagrams and generic discussion with verified model screenshots, input parameters and analysis results.

ABSTRACT

This report presents the workflow for modelling and analysing a G+1 reinforced concrete (RCC) structure using AutoCAD 2024 and STAAD.Pro. AutoCAD is used to prepare the building layout and structural drawings, while STAAD.Pro is used to create an analytical representation of the frame and evaluate the response of beams, columns and a cantilever member under gravity and seismic loading.

The workflow includes defining the structural grid and storey levels, drafting plans and elevations, idealising beams and columns as analytical members, assigning material and cross-sectional properties, applying support conditions, and defining load cases.

Seismic loading is considered as a lateral action in accordance with the applicable design code
and project-specific seismic parameters. Analysis outputs of interest include support reactions, member forces, storey displacements, drift and overall stability indicator

The project emphasises that reliable results depend on a consistent geometry, realistic boundary conditions, correct load paths and appropriate code parameters. Since detailed dimensions and numerical analysis output were not supplied for this report, no member capacity or compliance claim is made. The included figures are conceptual aids to communicate the modelling approach.

Keywords

G+1 building; reinforced concrete; AutoCAD 2024; STAAD.Pro; beam; column; cantilever; seismic loading; structural analysis.

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Project summary

Software: AutoCAD 2024 for drafting; STAAD.Pro for structural analysis. Structural system: G+1 reinforced concrete frame. Main members considered: beams, columns and a cantilever. Loading focus: gravity loading and seismic lateral loading, subject to project-specific code parameters.

1. INTRODUCTION AND PROJECT OVERVIEW

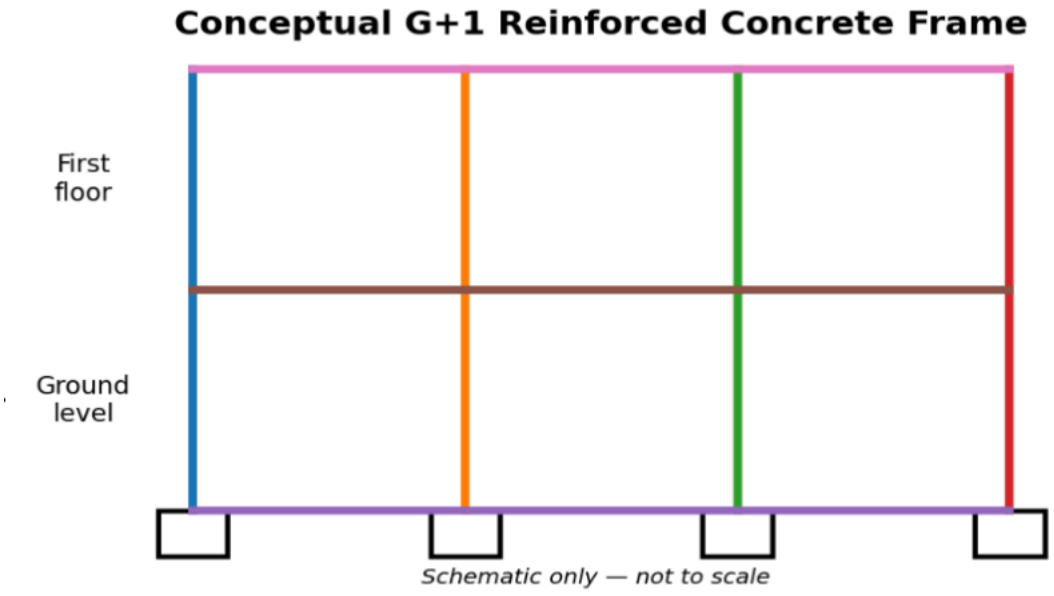
A G+1 building consists of a ground storey and one storey above it, typically supported by a system of slabs, beams, columns and foundations. In a reinforced concrete frame, slabs transfer floor loads to beams; beams transfer loads to columns; and columns transfer the accumulated actions to the foundation. The frame must also resist lateral actions, including those caused by earthquakes.

Earthquake loading differs from gravity loading because it produces inertia forces associated with the mass of the structure and can reverse direction. The distribution of stiffness, mass, member dimensions, connection behaviour and support conditions influences how lateral forces are carried through the frame.

A structural analysis model helps the engineer estimate member actions and displacements before design checks are completed.

AutoCAD 2024 is used for accurate 2D drafting and communication of the structural layout. STAAD.Pro presents the structure as a numerical model composed of nodes, members, supports, material properties and load cases.

The two tools serve complementary purposes: drawings describe geometry and detailing, while the analytical model estimates structural response.



2. OBJECTIVES AND SCOPE OF WORK

Objectives

· Prepare a clear structural layout and relevant drawings using AutoCAD 2024.

• Develop an analytical model of a G+1 reinforced concrete frame in STAAD.Pro.

• Represent beams, columns and a cantilever member with suitable geometry and properties.

• Define support conditions and load cases for gravity and seismic actions.

• Review member forces, support reactions and lateral displacement outputs.

• Understand how seismic loading affects the design and interpretation of structural response.

Scope

The scope covers the modelling and analysis workflow, software practices and the interpretation of key outputs.

Detailed reinforcement design, foundation design, construction detailing and final code compliance checks

require complete project inputs and are outside the scope of the information currently provided.

Information required for a final project-specific report

• Building plan dimensions, grid spacing, storey heights and cantilever projection.

• Beam and column sizes, slab thickness, concrete grade and reinforcement grade.

• Dead, live, wall and other applicable loads.

• Seismic zone, importance factor, response reduction factor, soil type and damping assumptions as required by

the applicable code.

• STAAD.Pro input file and analysis output, including displacement and member-force tables.

3.STRUCTURAL SYSTEM AND COMPONENTS

The structural system is idealised as a reinforced concrete moment-resisting frame. The actual structural arrangement must be taken from the approved plan and should preserve the intended alignment of columns, beams and floor levels. The following components are central to the analysis.

Beams

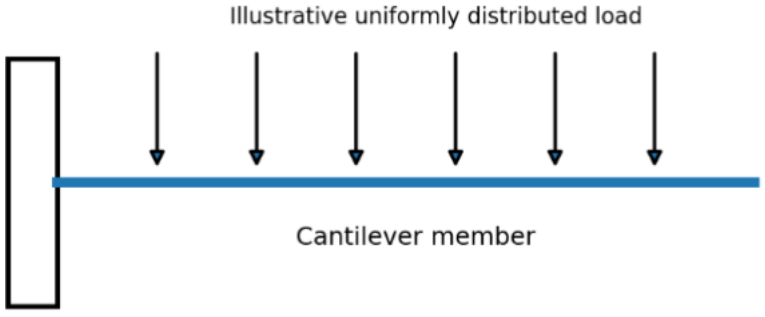
Beams primarily resist bending moments and shear forces arising from floor loads and frame action. Their end connectivity, spans, stiffness and continuity affect force distribution. Beam results should be reviewed for governing positive and negative bending moments, shear forces and deflection.

Columns

Columns carry axial forces together with bending moments and shear forces. Under seismic loading, frame columns can experience significant combined axial-flexural actions.Column checks should
consider the governing load combinations, slenderness where relevant, and applicable ductile detailing provisions.

Cantilever member

A cantilever is fixed or restrained at one end and projects without support at the other. It commonly develops maximum bending moment and shear near the restrained end.
Its model must represent the actual support



applied loads

Figure 2. Conceptual cantilever member under a distributed load.

4. METHODOLOGY AND MODELLING WORKFLOW

The workflow is organised so that geometry and design assumptions are established before analysis. Each stage should be checked before moving to the next, because errors in node connectivity, units or support conditions can lead to misleading results.



Figure 3. End-to-end workflow from drafting to analysis review.

Step-by-step procedure

- Review the architectural plan and establish column grid lines, spans and storey elevations.
- Prepare dimensioned plans and elevations in AutoCAD 2024; maintain consistent units and Layer naming
- Create the analytical geometry in STAAD.Pro using nodes and members, checking connectivity and member approved design assumptions.
- Define support conditions, self-weight and other gravity loads; confirm that loads act
- Create seismic load cases using the applicable code provisions and the required site- specific parameters.
- Run the analysis, inspect warnings and instability messages, then review reactions displacements and member forces
- Document the model assumptions, output tables, checks and any revisions made

5. DRAFTING IN AUTOCAD 2024

AutoCAD 2024 is used to communicate the structural geometry through accurate 2D drawings. A consistent drawing setup improves readability and reduces the possibility of transferring incorrect dimensions into the analytical model.

Recommended drafting sequence

- Set drawing units and confirm the intended unit system before creating geometry.
- Create separate layers for grids, columns, beams, slab edges, dimensions, text and annotations.
- Draw grid lines and label axes consistently; place columns at the intended grid intersections.
- Draw beam centre-lines or outlines and indicate slab boundaries and cantilever projections.
- Prepare floor plans and elevations that show storey heights, spans and member labels.
- Add dimensions, notes, a title block and drawing scale; check that the plan and elevation agree

Quality checks

Check that dimensions close correctly, columns align vertically between storeys unless a transfer system is intended, beam ends meet column locations, and the cantilever length matches the intended layout.

Drawing dimensions should be treated as the source of geometry, not inferred from an image or screen scale.

Recommended evidence for a final submission: a screenshot of the AutoCAD floor plan, an elevation, and a close-up showing dimensions and layers.

The schematic illustrations in this report are not a substitute for actual AutoCAD drawings.

6. ANALYTICAL MODELLING IN STAAD.Pro

STAAD.Pro is used to represent the building as an analytical frame and to solve for structural response under defined loading conditions. The model should represent the intended load path and relative stiffness of the structure.

Model setup

• Create nodes at beam-column intersections and at changes in member geometry.

• Connect nodes with beam and column members; verify that members are continuous where intended.

• Assign material properties and section sizes consistent with the project design basis.

• Specify supports to reflect the assumed foundation restraint; document any simplifications.

• Assign self-weight and additional loads to the relevant members or floor system representation.

• Define load combinations and seismic cases in accordance with the governing design code.

• Perform analysis and resolve modelling warnings before relying on results.

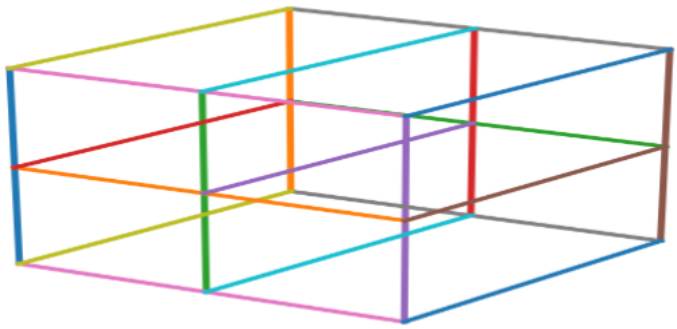
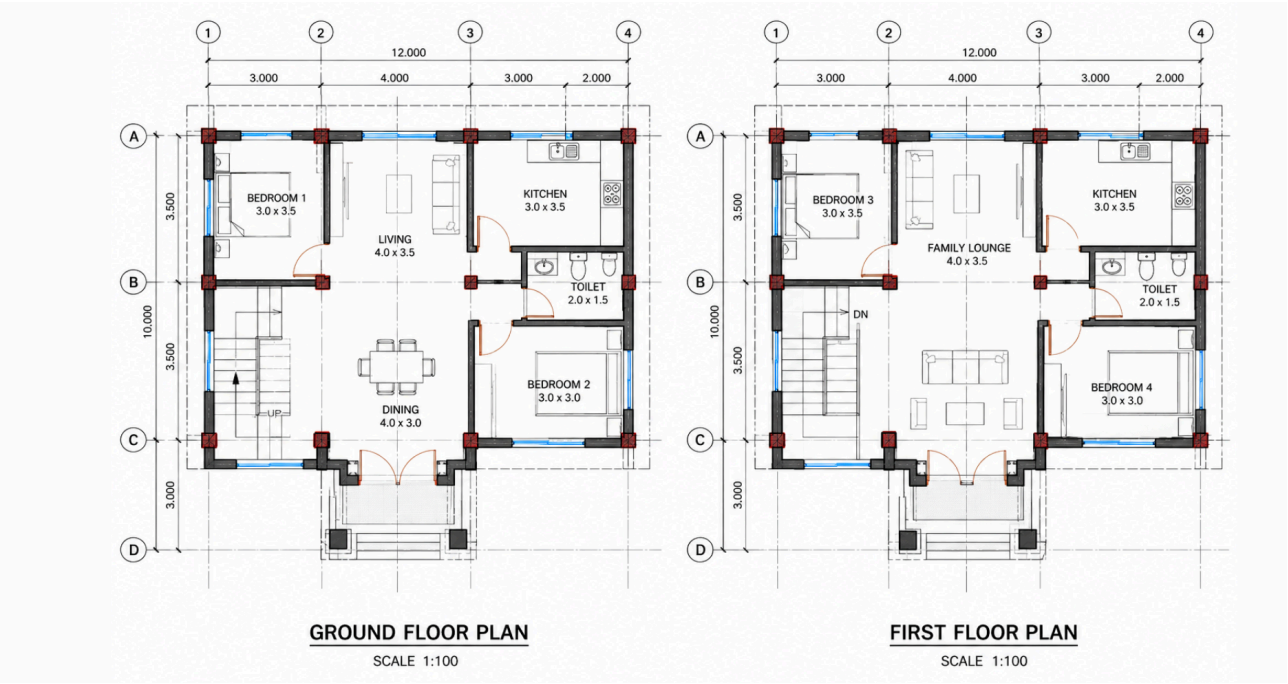


Figure 4. Illustrative 3D frame idealisation; actual model geometry may differ.

Important: a 3D rendering is only a visual check. Analytical validity also requires correct member connectivity, releases, offsets, units, boundary conditions, mass representation and load definitions.



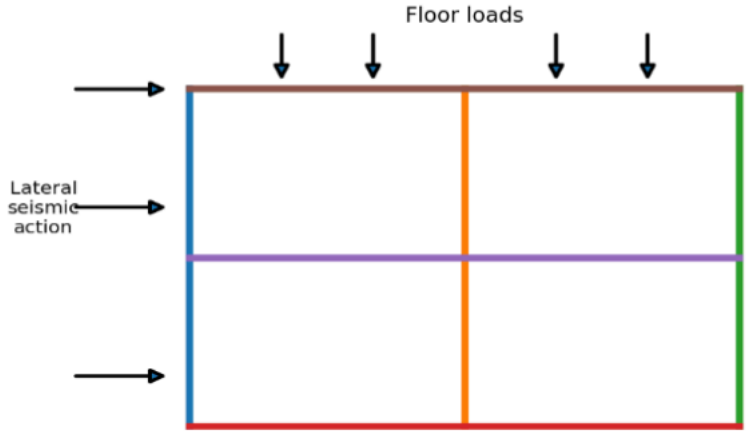
7. LOADING AND SEISMIC CONSIDERATIONS

The analysis should include all relevant actions identified by the design basis. Typical gravity actions include structural self-weight, floor finishes, partition or wall loads and imposed/live loads. Other actions may apply depending on the building use and project requirements.

Seismic loading

Seismic actions are generally represented by lateral forces or a code-defined analysis procedure.

The appropriate method depends on the building characteristics and the governing standard. Inputs commonly include seismic zone or hazard level, importance factor, response reduction factor, soil/site class, damping, seismic weight and the direction of earthquake action. These values must be selected for the actual site and occupancy; they are not assumed in this report.



The seismic load path transfers lateral inertia forces from floor levels through the frame to the supports and foundation.Review both principal horizontal directions where applicable, and consider accidental, torsion, orthogonal effects and load combinations where required by the relevant code.

Code and safety note

Use the current applicable Indian Standards and the project's approved design criteria. The specific seismic standard and edition should be confirmed by the supervising engineer. This report does not certify code compliance or establish safe member sizes.

8. ANALYSIS OF MEMBERS AND RESULTS REVIEW

Beams

Review bending moment and shear-force diagrams, governing load combinations, deflection and the locations of peak actions. Verify that end moments and shear forces are consistent with the expected continuity and loading.

Design checks should follow the relevant reinforced concrete code.

Columns

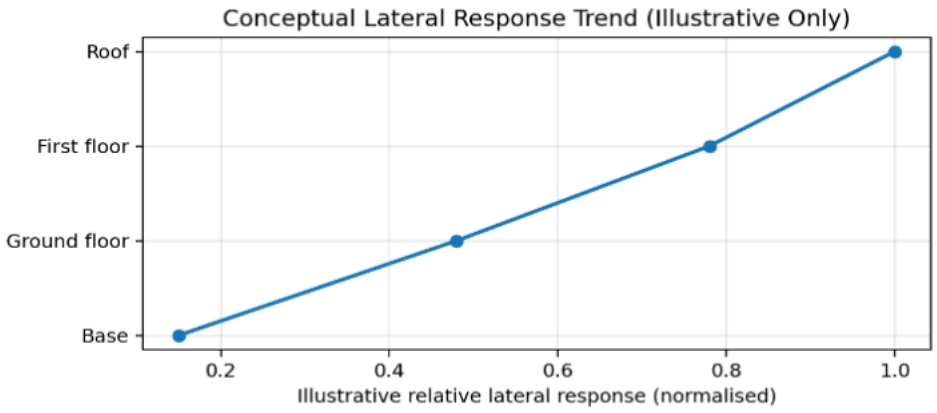
Review axial force, biaxial or uniaxial bending as applicable, shear and storey-level displacement. Check that the force combinations used for design include the required seismic effects and that the detailing approach is compatible with the adopted seismic design philosophy.

Cantilever

Check the maximum moment and shear at the fixed end, tip displacement and the effects of any concentrated or distributed loads. Confirm that the support idealisation represents the real structural connection.

Output checklist

The final model report should include: maximum displacement by storey; storey drift where required; support
reactions; governing beam moments and shears; governing column axial force and moments; cantilever tip displacement and fixed-end actions; analysis warnings; and the load combination controlling each reported value.



Figure

9. LEARNING OUTCOMES

• Improved understanding of the relationship between structural drawings and analytical models.

• Developed awareness of grid layout, member connectivity, dimensions and drawing organisation in AutoCAD

2024.

• Learned the main steps involved in building a frame model in STAAD.Pro.

• Understood how beams, columns and cantilever members respond to different load effects.

• Recognised the importance of seismic parameters, load combinations and realistic boundary conditions.

• Learned to review analysis warnings and interpret reactions, member forces and displacement outputs.

10. CONCLUSION AND RECOMMENDATIONS

The project outlines the modelling and structural analysis workflow for a G+1 reinforced concrete structure using AutoCAD 2024 and STAAD.Pro, with particular attention to beams, columns and a cantilever member under seismic loading.

The process links accurate drafting, analytical idealisation, assignment of properties and loads, seismic case definition, analysis and interpretation of outputs.

The reliability of the final design depends on project-specific geometry, material strengths, member sizes, loading, site seismic parameters and a verified analytical model.

The next step should be to insert the actual AutoCAD drawings and STAAD.Pro screenshots, provide the governing input parameters, and report numerical analysis results with the relevant design checks.

The model should be reviewed by a qualified structural engineer before any construction use.

11. REFERENCES

1. Bureau of Indian Standards (BIS), IS 456: Plain and Reinforced Concrete — Code of Practice (confirm current applicable edition).

2. Bureau of Indian Standards (BIS), IS 1893 (Part 1): Criteria for Earthquake Resistant Design of Structures (confirm current

applicable edition).

3. Bureau of Indian Standards (BIS), IS 13920: Ductile Design and Detailing of Reinforced Concrete Structures Subjected to Seismic

Forces (confirm current applicable edition).

4. Bentley Systems, STAAD.Pro product documentation and help resources.

5. Autodesk, AutoCAD 2024 documentation and user guide.